ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C1 · MATTER

States and substances

★☆☆
Grade 4

Learning goals

- Sort solids, liquids and gases.
- Name changes of state.
- Sort elements, compounds and mixtures.

1 Sort the materials. [9]

SORT THESE

rice grain **milk** **ice** **stone** **cooking gas** **air in a balloon** **steam** **rain water** **mustard oil**

Solid	Liquid	Gas
stone	milk	steam
ice	mustard oil	air in a balloon
rice grain	rain water	cooking gas

2 Name the change of state. [4]

What happens	Change of state
ice cream drips on a hot day	melting
water is put in a freezer and becomes ice	freezing
a puddle disappears in the sun	evaporating
drops form on a cold glass of water	condensing

3 Draw the particles in each state. [3]

solid
neat rows, touching

liquid
touching but jumbled

gas
few, far apart

4 **Sort the substances.** [9]

SORT THESE

gold **water** **oxygen** **iron** **soil** **sea water** **air** **carbon dioxide** **salt**

Element	Compound	Mixture
iron	water	air
oxygen	salt	sea water
gold	carbon dioxide	soil

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · C1 · MATTER

Explaining matter

★★☆
Grade 4

Learning goals

- Explain properties using particles.
- Use melting points.

1 **Use the particle model to explain why:** [4]

a) a liquid takes the shape of its container.

Its particles can slide past each other, so the liquid flows to fill the bottom of the container.

b) you can squash a gas but not a solid.

Gas particles are far apart with lots of space between them; solid particles are already packed tightly.

2 **Chocolate melts at 34 °C and candle wax melts at 60 °C. On a 38 °C summer day, which one melts in your bag? Why?** [2]

The chocolate, because 38 °C is above its melting point, but below the melting point of wax.

3 **Rumi says, "Salt is an element because it is pure." Is she right? Explain.** [2]

No. Salt is pure, but it is a compound: two kinds of atom (sodium and chlorine) joined together.

4 **Why is it useful that iron has a very high melting point?** [2]

Iron pots and stoves do not melt when they are heated on a fire.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · C1 · MATTER

The melting race

★★☆
Grade 4

Learning goals

- Plan a fair test.
- Time and record results.
- Link results to melting points.

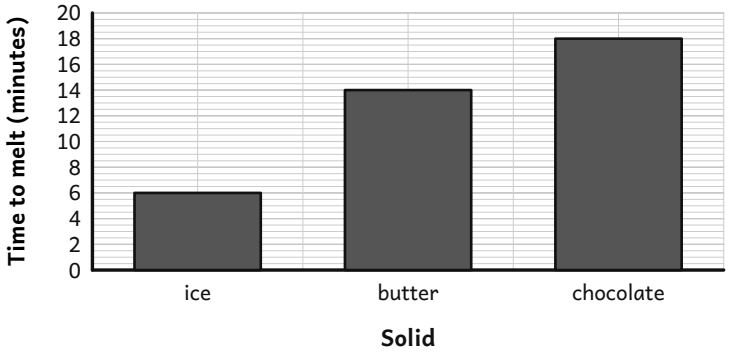
Class 4B put a 1 cm cube of four solids on four identical metal lids in the same sunny place, and timed how long each took to melt completely.

1 **Complete the plan.** [4]

Change	the type of solid
Measure	time to melt (minutes)
Keep the same	size of the cube, type of lid, place in the sun, start time
Prediction	e.g. ice will melt first because it has the lowest melting point

2 **Draw a bar chart of the results.** [3]

Solid	ice	butter	chocolate	candle wax
Time to melt (minutes)	6	14	18	did not melt



3 **Why did the candle wax not melt?** [1]

The lid did not get as hot as 60 °C, the melting point of wax.

4 **Describe the pattern between melting point and time to melt.** [2]

The lower the melting point, the faster the solid melted.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C1 · MATTER

Ice for the fishing boats

★★★
Grade 4

Learning goals

- Apply changes of state to a real job.
- Use numbers.
- Think about energy and waste.

Science · Economics · Maths · Health

Before a fishing trawler leaves Chattogram or Cox's Bazar for the Bay of Bengal, workers load it with tonnes of **crushed ice**. A trip can last a week or more. The fishers pack each catch of hilsa and other fish in ice so it stays fresh until they return.

The ice is made in **ice factories** near the harbour. Machines cool water below **0 °C** until it **freezes** into big blocks, which are crushed before loading. As the ice slowly **melts**, it takes heat from the fish and keeps them cold. Without ice, fish start to go bad in a few hours in the heat.

Based on information about fish preservation in Bangladesh's fishing industry. Numbers below are examples.

1 **Which changes of state happen in the story?** [2]

Where	Change of state
in the ice factory	freezing (liquid → solid)
in the boat's fish hold	melting (solid → liquid)

2 **Why is crushed ice better than one big block for keeping fish cold?** [2]

Small pieces fit closely around every fish and touch more of its surface, so the fish cool faster.

3 **Maths. A boat needs 2 kg of ice for every 1 kg of fish. It hopes to catch 500 kg. How much ice does it need?** [2]

500 × 2 = 1000 kg (1 tonne)

4 **Making ice uses a lot of electricity. Suggest one way to waste less ice.** [2]

Use insulated boxes in the boat (like a cool box), keep the hold closed and in the shade.

5 **Why is it important for people's health that fish are kept cold?** [1]

Germs grow fast in warm fish and can make people ill; cold slows them down.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · C1 · MATTER

Show what you know

★★★
Grade 4

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

W	R	N	B	I	Y	Y	W	S	L
L	I	Q	U	I	D	R	U	O	M
C	N	N	P	F	B	P	W	L	J
M	A	T	T	E	R	O	A	I	S
U	V	M	L	G	C	E	I	D	B
E	L	E	M	E	N	T	E	L	O
V	R	L	S	M	J	K	O	Z	K
U	I	T	S	S	A	F	G	I	E
M	I	X	T	U	R	E	L	A	R
W	M	V	K	W	U	C	G	U	S

boil element freeze gas liquid matter melt mixture solid

2 At what temperature does pure water boil? [1]

- ○ 0 °C
- ○ 50 °C
- ● **100 °C**
- ○ 34 °C

3 Which is a compound? [1]

- ○ oxygen
- ● **water**
- ○ gold
- ○ air

4 Draw a diagram to show what happens to the particles when ice melts. [3]

Ice melting

Particles in neat rows (solid) → particles touching but jumbled (liquid), with an arrow labelled "heat".

5 **Challenge. A glass of iced water gets wet on the outside. Where does the water come from? Explain using states of matter.** [2]

Water vapour (a gas) in the air touches the cold glass, cools and condenses into liquid water drops.